What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Cholinergics and Anti-cholinergics –1/2

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Adapted from: Cholinomimetic Agents

Listen to this Brick

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Classify cholinergic receptors based on their G protein class and explain the associated G-protein-linked second messenger pathways.
2. Describe the tissue expression pattern of the cholinergic receptors.
3. Predict the o ivation and inhibition of cholinergic receptors based on the tissue expression patterns.
4. Explain the therapeutic uses of cholinergics and anticholinergics based on their target organs and receptor selectivity.
5. Describe the potential adverse effects of cholinergics and anticholinergics

CASE CONNECTION

NK is a 76-year-old male with severe lower abdominal pain. "I haven't peed in several days, and I feel like I might explode," he says. He has never had trouble like this before and is generally healthy. He has been taking an over-the-counter allergy pill recently because of a runny nose. On exam, you note a markedly distended bladder; you quickly and carefully empty his bladder by catheterization. He immediately feels better and thanks you profusely. "I don't want to go through that ever again," NK says. "What caused this?"

What will you tell NK about the mechanisms behind urinary retention? Consider your answer as you read, and we'll revisit NK at the end of both the Cholinergics and Anticholinergics bricks.

GO TO CONCLUSION ↓

WHAT ARE CHOLINERGIC AGENTS? (LO#1)

Cholinergics (or chol ics (-mimetic) acetylcholine (cholino-). As you will recall from the ANS brick, acetylcholine (ACh) is the primary distal neurotransmitter of the parasympathetic nervous system (PNS), activating the cholinergic (or Muscarinic) receptors. Cholinergics are divided into **direct** and **indirect-acting** cholinergics (discussed in a Week 4, Day 1 brick).

- **Direct-acting** cholinomimetic agents bind to acetylcholine receptors and cause activation of the PNS.
- **Indirect-acting** cholinomimetic agents are also called **cholinesterase. inhibitors**, since they inhibit the breakdown of acetylcholine by the enzyme cholinesterase.

What Happens When Muscarinic Receptors Are Activated? (LO#2)

First let's look at the normal effects of muscarinic receptor activation. In the ANS vasopressor/inotropes brick, we discussed the five muscarinic receptor subtypes: M1-M5, with M1-M3 being the most important clinically. Take a minute to attempt the active learning question below. It should help you recall the Muscarinic receptor activity and will help you understand the cholinergic mechanism of action.

ACTIVE LEARNING QUESTION

Match the Muscarinic and Nicotinic receptor to its GPCR subtype or ion-chanel and downstream signaling activity

M2	✓ Gi--> decrease cAMP
M1 and M3	✓ Gq --> increase IP3 and Ca release
Nm	✓ Na/K channel activation

Explanation

M1 -M3 are Muscarinic cholinergic receptors, Nm is Nicotinic muscarinic receptor. Nm is a ligand-gated ion channel which is typically conduction Na or K. M1-M3 are muscarinic receptors that are GPCRs. M1 and M3 are of Gq subtype and hence increase intracellular Ca levels with IP3 receptor activation. M2 is a Gi subtype and is associated decrease in function (for example, M2 is expressed in the heart decreasing heart rate)

Additionally, in the ANS review and KIFS brick we have discussed the effects on cholinergic (or PSNS) activation on various organs. A recap of the effects on organs is listed below:

- Cardiovascular: Slowing the pulse rate and impulse conduction through the heart nodes. [M2]
- Endocrine: Increased secretions from sweat, salivary, lacrimal, and nasopharyngeal
- Eye: Contraction of the sphincter and ciliary muscles of the eye, leading to miosis (pupil contraction). [M3]
- Gastrointestinal: Increased gastric motility, increase gastric secretions, relaxed internal anal sphincter. [M3]
- Musculoskeletal: Increased muscle contraction.
- Neurologic: Improved cognition, memory, and learning.
- Respiratory: Bronchoconstriction and increased bronchial gland secretions.
- Urinary: contraction of the bladder detrusor muscle, relaxation of the internal urethral sphincter. [M3]

Which muscarinic receptor subtype is primarily responsible for slowing the heart rate?

How do direct–acting cholinergics work? (LO#3)

Direct-acting choline eceptor function and produce similar cholinergic effects. As seen in the figure below, these agents are capable of producing the same effects in the organs that we discussed in the previous section (Figure 1). The effects of these agents are largely dependent on the route of administration and formulation.

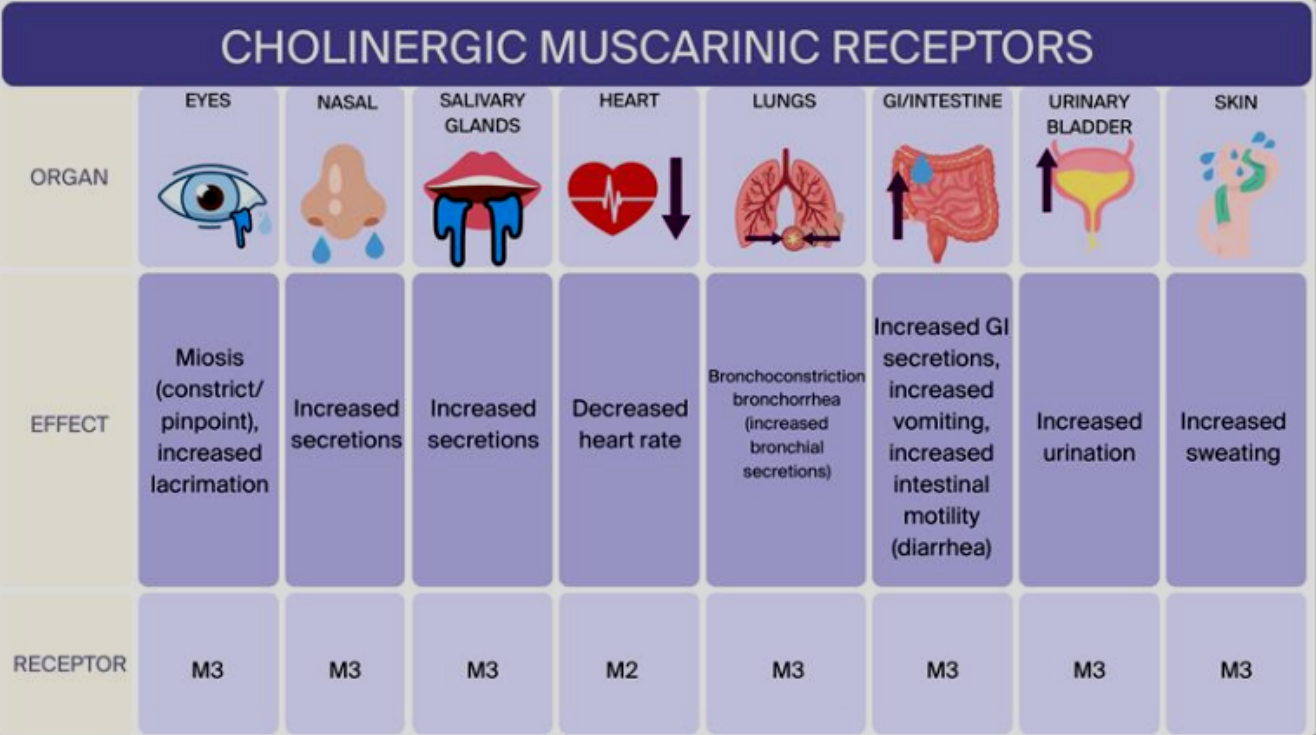
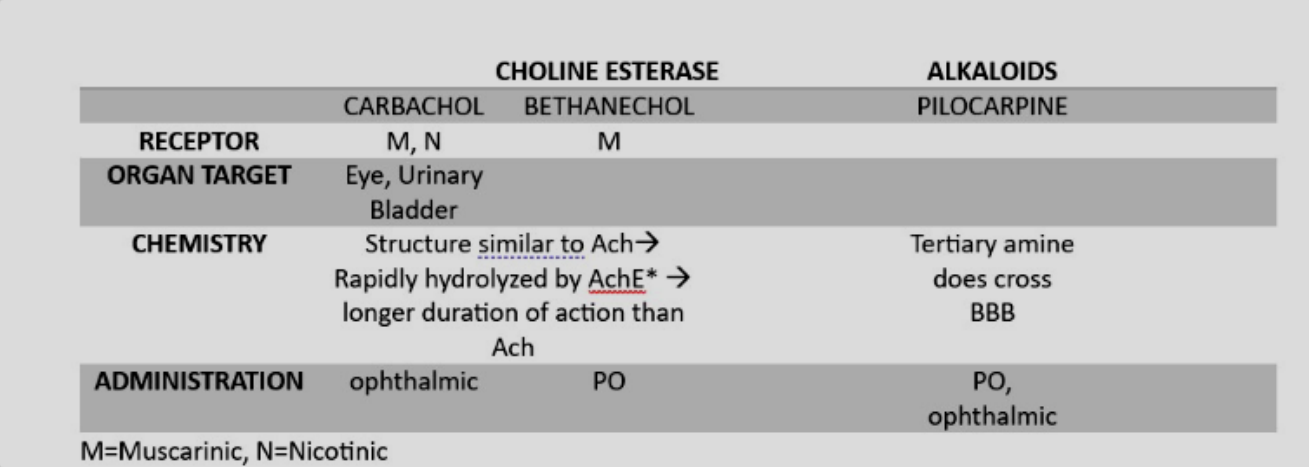

CHOLINERGIC MUSCARINIC RECEPTORS
ORGAN
EFFECT
RECEPTOR
EYES
NASAL
SALIVARY GLANDS
HEART
LUNGS
GI/INTESTINE
URINARY BLADDER
SKIN
Miosis (constrict/ pinpoint), increased lacrimation
Increased secretions
Increased secretions
Decreased heart rate
Bronchoconstriction bronchorrhea (increased bronchial secretions)
Increased GI secretions, increased vomiting, increased intestinal motility (diarrhea)
Increased urination
Increased sweating
M3
M3
M3
M2
M3
M3
M3
M3

Figure 1

What are the phy SNS in the lungs?

What is the effect of cholinergics on the urinary system?

We know that ACh is rapidly hydrolyzed by AChE(Acetylcholinesterase). Since choline esters are esterified choline derivatives (modified side chains), they are hydrolyzed by AChE. However, the modification to the side chain makes them more resistant to hydrolysis (compared to ACh). This is why these agents have longer duration of action than ACh. The agents end with a suffix --chol (modified choline esters). Commonly used direct-acting cholinergics are listed in Table 1. Pilocarpin thus can cross the blood brain barrier (B



	CHOLINE ESTERASE		ALKALOIDS
	CARBACHOL	BETHANECHOL	PILOCARPINE
RECEPTOR	M, N	M	
ORGAN TARGET	Eye, Urinary Bladder		
CHEMISTRY	Structure similar to Ach→ Rapidly hydrolyzed by AchE* → longer duration of action than Ach		Tertiary amine does cross BBB
ADMINISTRATION	ophthalmic	PO	PO, ophthalmic

M=Muscarinic, N=Nicotinic

Characteristics of Direct-Acting Cholinergics

What Are Common Direct–Acting Cholinomimetics and Their Clinical Uses? (LO#4)

Because muscarinic receptors are so widely distributed, the direct agents have the ability to affect nearly every organ system and therefore can treat a wide range of conditions. The three commonly used main direct cholinergics agonist are:

- *Bethanechol* is used to treat urinary retention by causing bladder contractions and relaxing the urinary sphincter. It's also used to treat severe constipation and bowel ileus (slow peristaltic contractions) by stimulating gastric motility.
- *Carbachol* is used as eye drops in the management of glaucoma because they reduce resistance to aqueous humor outflow. These drugs cause miosis (small pupils) by contracting the pupillary sphincter muscle and adjusting the len iction of the ciliary muscle that controls the lens shape.
- *Pilocarpine* is also used as eye drops in the management of glaucoma because it reduces resistance to aqueous humor outflow. Pilocarpine is used for xerostomia following head and neck radiation treatment (oral formulation). Patients with Sjögren syndrome (autoimmune disorder) whose lacrimal and salivary functions are disrupted use pilocarpine to improve function.

CLINICAL CORRELATION

Note that, except for methacholine, these direct-acting drugs are resistant to breakdown by acetylcholinesterase, which can make them more potent and longer-acting agents than acetylcholine itself. Methacholine is indicated for testing bronchospastic activity as a diagnostic tool and has no other therapeutic use.

WHAT ARE POTENTIAL ADVERSE EFFECTS FROM CHOLINERGIC USE? (LO#5)

Adverse Effects. Most direct-acting cholinergics have few adverse effects unless a toxic dose has been administed. The effect of toxic overdose is discussed in the indirect-acting cholinergics brick. Use of Pilocarpine is associated with some adverse effects. Pilocarpine use can cause excess sweating. About 30-60% of patients complained of excess sweating.

CASE CONNECTION

BACK TO INTRODUCTION ↑

The conclusion to the case is described in the Cholinergics/Anticholinergics 2/2. However, pay attention to the signs and symptoms described by NK. These will help understand the case.

Summary

- Cholinomimetic agents mimic the action of acetylcholine, a primary neurotransmitter of the parasympathetic nervous system (PNS).
- There are two types of cholinomimetics: direct-acting agents that bind to acetylcholine receptors and indirect-acting agents that inhibit the breakdown of acetylcholine, both leading to PNS activation.
- Activation of muscarinic receptors in the PNS has various physiological effects, including slowing heart rate, increasing gland secretions, contracting pupils, increasing gastric motility, improving cognition, causing bronchoconstriction, and contracting the bladder.
- Direct-acting cholinomimetics like bethanechol, carbachol, and pilocarpine are used to treat conditions such as urinary retention, glaucoma, constipation, and Sjögren syndrome.
- Potential adverse effects of cholinomimetic use include excessive sweating, diarrhea, miosis, bronchoconstriction, bradycardia, and salivation.
- Cholinomimetic agents can be used to treat conditions like myasthenia gravis and to reverse the effects of nondepolarizing neuromuscular drugs in anesthesia.

Review Questions

1. Activation of muscarinic receptor in the heart leads to

A. ● decrease in heart rate

B. ○ increase in heart rate

C. ○ no change in heart rate

Hide Explanation

The correct answer is a decrease in heart rate (A). Activation of M2 receptors decreases heart rate. M2 receptor activation results in Gi type of GPCR decreasing cardiac contractility. An increase in heart rate (B) is incorrect since activation of M2 results in activation of the Gi pathway and not Gs, which can increase heart rate. No change in heart rate (C) is incorrect since activation of PSNS or muscarinic receptors results in a 'rest and digest' function, including a decrease in heart rate.

2. Which direct cholinergic agonist is used for xerostomia?

A. ○ Bethanechol

B. ○ Carbachol

C. ● Pilocarpine

D. ○ Acetylcho

Hide Explanation

The correct answer is Pilocarpine (C). Pilocarpine is used for xerostomia following head and neck radiation treatment and in patients with Sjögren syndrome. Bethanechol (A) is used for urinary retention and bowel ileus. Carbachol (B) is used for glaucoma. Ae (D) Acetylcholines are direct-acting cholinergic but very short-acting and not used frequently in practice.

3. Which direct-acting cholinergic is used for urinary retention?

A. ● Bethanechol

B. ○ Carbachol

C. ○ Pilocarpine

D. ○ Acetylcholine

Hide Explanation

The correct answer is Bethanechol (A). Bethanechol is used to treat urinary retention by causing bladder contractions and relaxing the urinary sphincter. Carbachol (B) and Pilocarpine (C) are primarily used in the management of glaucoma. Acetylcholine (D) is used for GI motility and does not cause xerostomia.

4. A cholinergic agent was developed that acts on the M2 receptors in the Lungs. This agent would cause

A. ○ Bronchodilation

B. ○ Bronchial dryness

C. ● Bronchoconstriction

Hide Explanation

The correct answer is Bronchoconstriction (C). This new agent will act on the M3 receptor to cause bronchoconstriction. Cholinergics will activate the PSNS tone. Thus bronchodilation (A) is incorrect. Activation of PSNS also increases secretions in the lungs, sweat glands, eyes, and nasal passages. Therefore, bronchial dryness (B) is incorrect.

5. What effect will a cholinergic agent likely have on GI/Intestinal motility?

A. ○ Decreased gastric motility

B. ○ Constipation

C. ○ Decreased gastric secretions

D. ● Diarrhea

Hide Explanation

The correct answer is Diarrhea (D). The cholinergic agent will increase gastric motility and cause diarrhea. Muscarinic receptor activation cause increase in gastric motility and secretions. Thus, choices (A) and (C) are incorrect. Since gastric motility is increased constipation (A), which is a decrease in gastric motility is incorrect.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Classify cholinergic receptors based on their G protein class and explain the associated G-protein-linked second messenger pathways.~~

02 ☑ ~~Describe the tissue expression pattern of the cholinergic receptors.~~

03 ☑ ~~Predict the organ-specific effects resulting from activation and inhibitio ssue expression pattern~~

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Objective Confidence: 5/5